



STA201 Assignment 1 (Spring 2023)

[Total Marks: 30]

Question 1

The following table shows some information on a variety of different vehicles. Using the Information given in table 1, answer question **1(A) – 1(C)**.

[3+3+2]

Table-1: Cars

Model	Number of gears	Size of Engine	Transmission	Vehicle Class	Highway Mileage	City Mileage	Model Year
Allion	6	3.4	Manual	Midsize Car	22	17	1997
Axio	7	2.5	Manual	Small Car	26	22	2000
Aqua Hybrid	8	3.5	Automatic	Small Car	30	24	2021
Corolla	5	3	Manual	Midsize Car	25	22	2001
Vezel	8	6.8	Automatic	Large Car	18	15	2022
Premio	6	3.3	Automatic	Midsize Car	22	17	2016
Harrier	7	4.2	Automatic	Large Car	18	15	2020
Naoh	7	5.4	Automatic	Large Car	20	16	2018

Question 1(A)

- (i) How many variables are listed in table 1?
- (ii) Classify the variables according to their types (Qualitative / Quantitative).
- (iii) Construct a frequency distribution table to represent the summary information of the Variable “**Vehicle Class**” and display the results in a pie chart.

Question 1(B)

Complete the following table and answer the questions:

Table 2: Frequency Distribution of Transmission by Number of Gears

Transmission	Number of Gears				Total
	5	6	7	8	
Automatic					
Manual					
Total					

- (i) What is the modal response for the variable “**Transmission**”? (Which has the highest frequency?)
- (ii) What proportion of vehicles have seven gears?
- (iii) What proportion of Automatic vehicles have eight gears?
- (iv) What proportion of vehicles with six gears are Manual?
- (v) Construct a side-by-side bar chart to represent the information given in table 2.

Question 1(C)

Complete the following table and answer the questions:

Table 3: Frequency distribution of Highway Mileage

Highway Mileage	Tally	Frequency	Relative frequency	Cumulative relative frequency
15 – 20				
20 – 25				
25 – 30				
30 – 35				

- (i) What proportion of vehicles have mileage between 20 and 30 Highway Mileage?
- (ii) What proportion of vehicles have mileage of at least 30 Highway Mileage?
- (iii) Construct a histogram to display the data represented in table 3.

Question 2

Recall that 65 purchasers have participated in a survey and have rated the XYZ-Box video game system. The composite ratings that have been obtained are as follows: [3+3]

39	38	40	40	40	46	43	38	44	44	44
45	42	42	47	46	45	41	43	46	44	42
38	46	45	44	41	45	40	36	48	44	47
42	44	44	43	43	46	43	44	44	46	43
42	40	42	45	39	43	44	44	41	39	45
41	39	46	45	43	47	41	45	45	41	

- (a) Construct a stem-and-leaf display for the 65 composite ratings.
- (b) If we consider a purchaser to be “very satisfied” if his or her composite score is at least 42, can we say that almost all purchasers of the XYZ-Box video game system are “very satisfied”. Describe in your own words.

Question 3

You are given the following table where the number of stocks falling in the particular price categories in the Chicago Stock Exchange (CSE): [2+2+2]

Stock Prices (\$)	No. of Stocks
50 – 60	8
60 – 70	$\sqrt{144}$
70 – 80	19
80 – 90	24
90 – 100	$\sqrt{25} + X + Y$

Here, **X** & **Y** are the **last two digits** of your student ID (i.e., 191000**12**, 201234**78**, etc.). [If your ID is 191000**12** then, the No. of Stocks in the last row (90 – 100) will be $\sqrt{25} + \textcolor{blue}{1} + \textcolor{red}{2} = 8$]. Estimate the following:

- a) Find:
 - i. Mean
 - ii. Median
 - iii. Mode

Question 4

A group of colleagues is going to Safari Park for a day. They arranged their trip into three halves and planned to travel at a speed of 68 kilometers per hour on their journey to their goal. Their top speeds were 72 km/h and 88 km/h for the first and second parts, respectively. If they are to meet their intended average speed, what speed should they maintain throughout the third part of their journey? [4]

Question 5

Suppose, you spent Tk. 3,500,000 to buy a new car. After the first year, the car depreciates by 40%, 20% after the second year, and 10% after the third year. [3+3]

- (a) What is the average rate of depreciation per year after three years?
- (b) What will be the value of the car after three years of use?